

LECTURE 4

Manometry and Hydrostatic Forces

Reference: Munson, Rothmayer, Okiishi & Huebsch, Fundamentals of Fluid Mechanics, Chapter 2
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Manometry

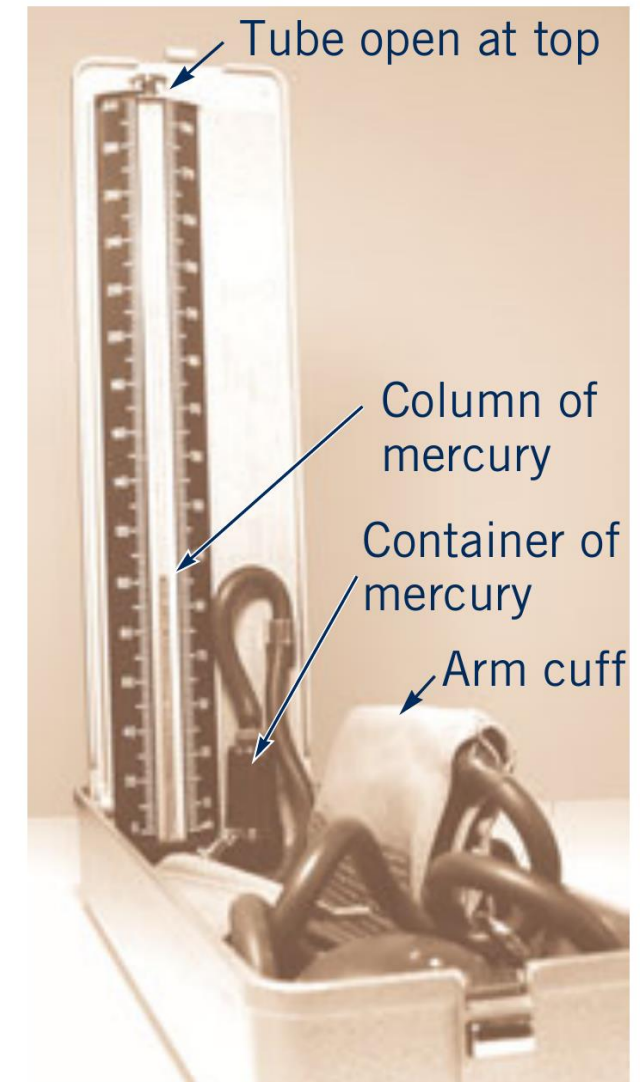
Manometers use vertical or inclined liquid columns to measure pressure.

A manometer measures pressure with a column of liquid at rest, so the only equation needed is $p = \gamma h + p_0$, applied step by step along the tube.

Three types are used:

- the piezometer tube
- the U-tube manometer and
- the inclined-tube manometer

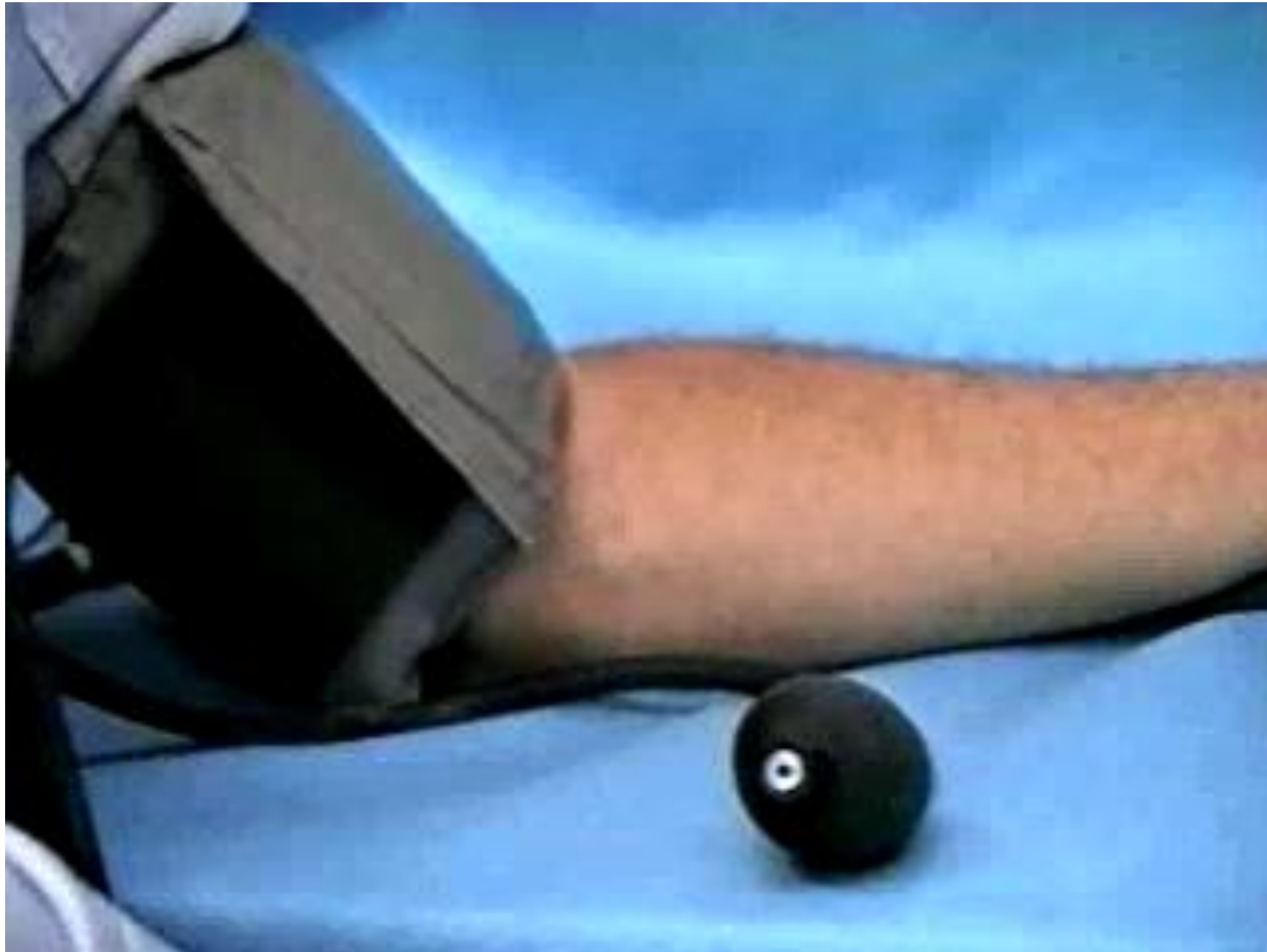
A **sphygmomanometer** is the traditional instrument used to measure blood pressure.



The sphygmomanometer works on this principle.

Measuring blood pressure

Blood pressure is quoted as the ratio of systolic to diastolic pressure, in millimetres of mercury — a pressure head, read directly off a manometer column.



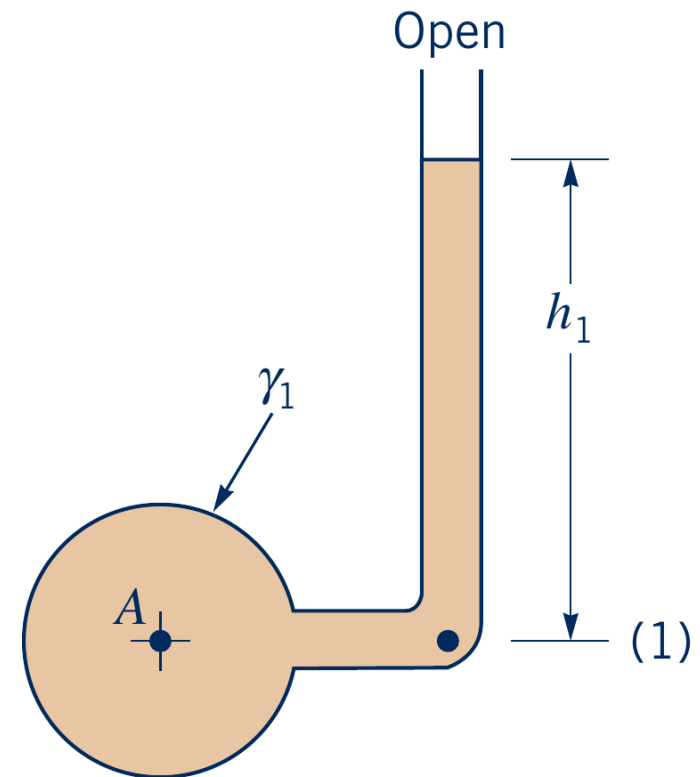
Blood pressure measurement with a sphygmomanometer.

Piezometer tube manometer

The piezometer tube is the simplest:

It is a vertical tube, open at the top, attached to the container A. Since point (1) and point A are at the same elevation, and the top is open:

$$p_A = \gamma_1 h_1$$



It is an accurate pressure-measuring device, but it works:

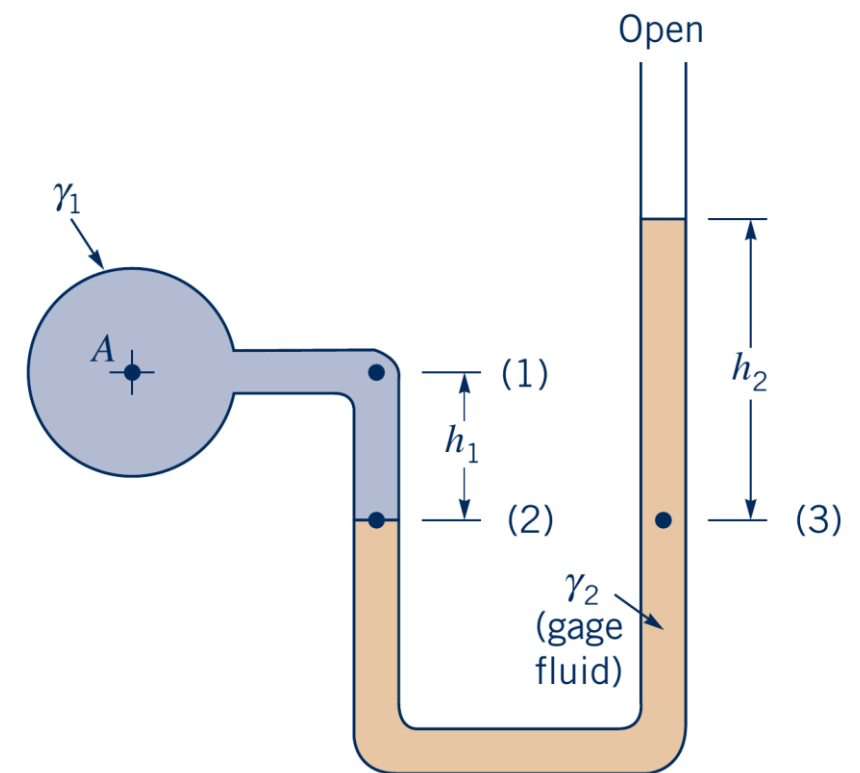
- Only if the pressure is above atmospheric
- Only if the required column height is reasonable (small p).
- Only if the fluid in the container is a liquid

The U-tube manometer

The fluid in the tube is the gage fluid. Start at one end of the system and work around to the other, adding γh when moving down and subtracting γh when moving up:

$$p_A + \gamma_1 h_1 - \gamma_2 h_2 = 0$$

$$p_A = \gamma_2 h_2 - \gamma_1 h_1$$



Points (2) and (3) are at the same elevation in one continuous mass of fluid, so their pressures are equal.

Points (1) and (2) are not: they lie in different fluids, so we cannot jump across.

If A contains a gas, the column $\gamma_1 h_1$ is negligible against the gage-fluid term:

$$p_A \simeq \gamma_2 h_2$$

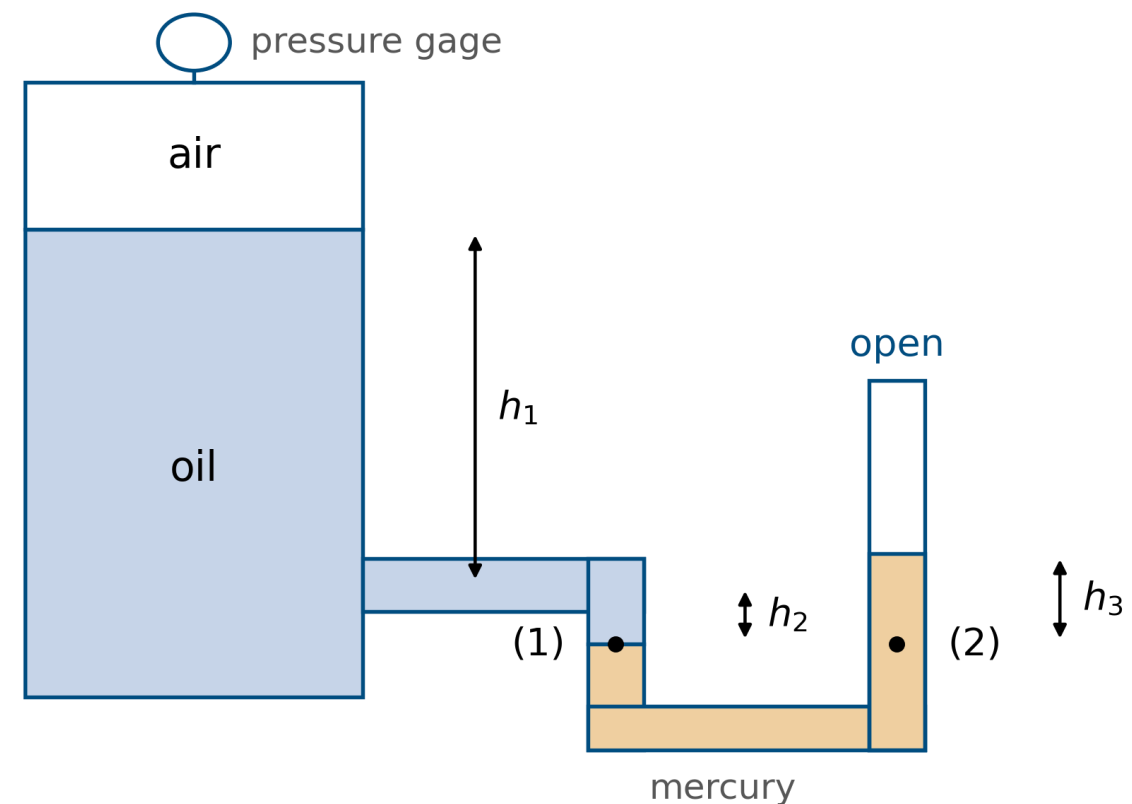
Example — closed tank with a U-tube manometer

A closed tank holds compressed air above oil of SG = 0.90. A U-tube manometer using mercury, SG = 13.6, is connected to it. The column heights are $h_1 = 900$ mm, $h_2 = 150$ mm and $h_3 = 230$ mm. Determine the pressure the gage on the tank should read.

Start at the air–oil interface and walk to the open end, where the pressure is zero:

$$p_{\text{air}} + \gamma_{\text{oil}}(h_1 + h_2) - \gamma_{\text{Hg}}h_3 = 0$$

$$\begin{aligned} p_{\text{air}} &= (13.6)(9.80)(0.230) - (0.90)(9.80)(1.050) \\ &= 21.4 \text{ kPa} \end{aligned}$$



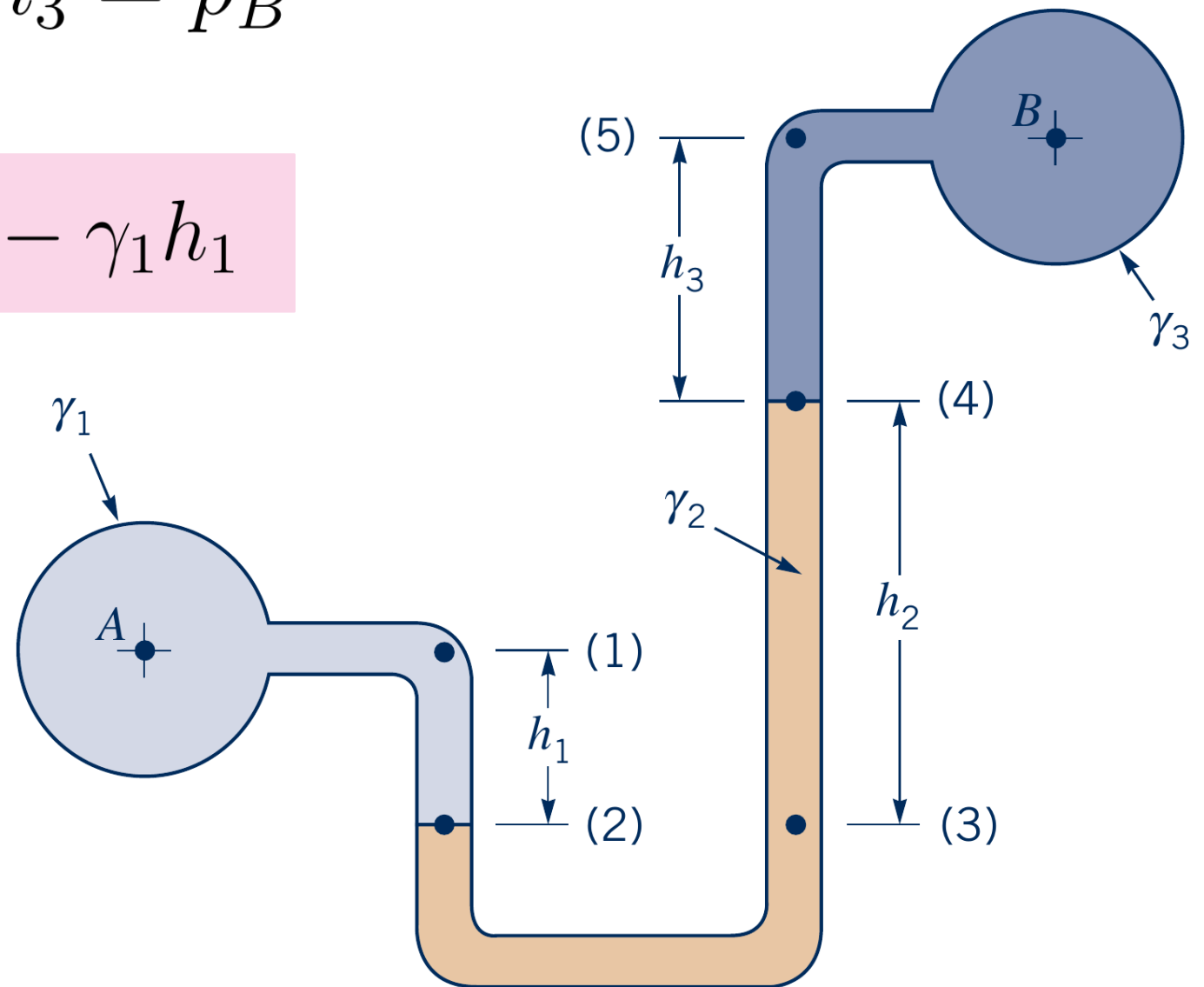
The air pressure depends on the mercury column and on the depth of the oil, both in the tank and in the tube. It is not the mercury alone.

The differential U-tube manometer

The same instrument measures the difference in pressure between two containers (A and B), or two points in a system. Walk from A round to B:

$$p_A + \gamma_1 h_1 - \gamma_2 h_2 - \gamma_3 h_3 = p_B$$

$$p_A - p_B = \gamma_2 h_2 + \gamma_3 h_3 - \gamma_1 h_1$$



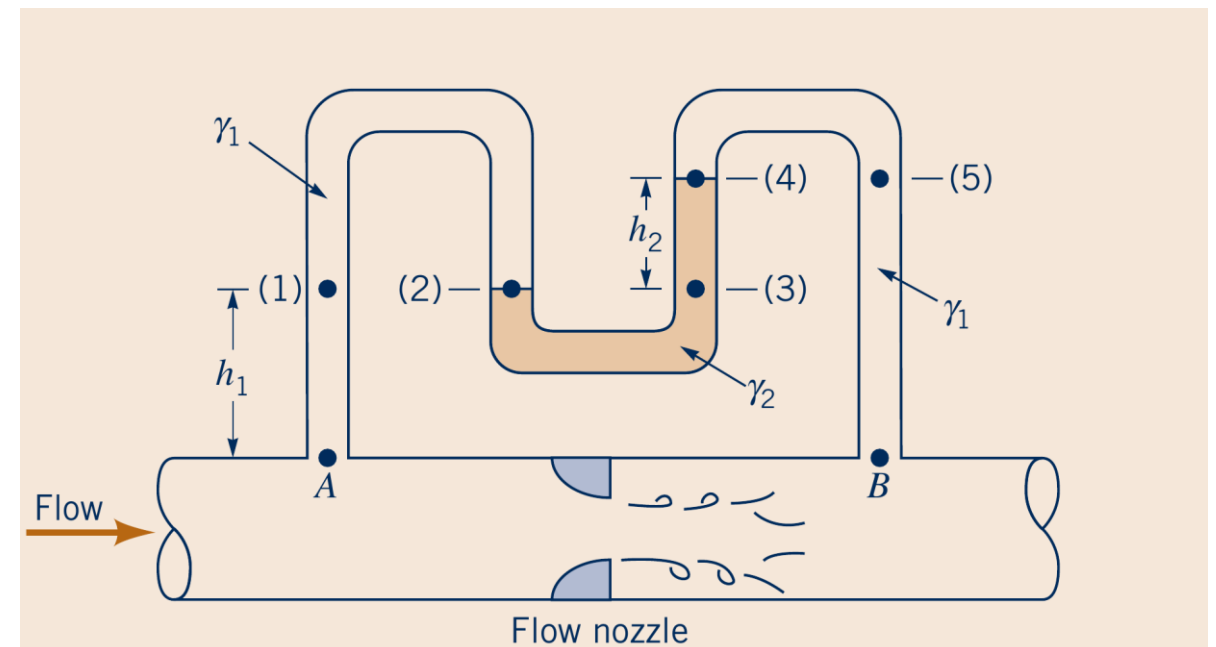
Starting at B and working to A gives the same result.

➡ Manometers are often used to measure a difference in pressure

Example — pressure drop across a flow nozzle

A flow nozzle in a horizontal pipe creates a pressure drop $p_A - p_B$ that is related to the flow rate. The drop is read on a differential **U-tube manometer**. Find an expression for $p_A - p_B$, and evaluate it for $\gamma_1 = 9.80 \text{ kN/m}^3$, $\gamma_2 = 15.6 \text{ kN/m}^3$, $h_1 = 1.0 \text{ m}$ and $h_2 = 0.5 \text{ m}$.

$$p_A - p_B = h_2 (\gamma_2 - \gamma_1)$$



$$p_A - p_B = (0.5 \text{ m})(15.6 - 9.80 \text{ kN/m}^3) = 2.90 \text{ kPa}$$

Only the differential reading h_2 matters. The manometer may sit 0.5 m or 5 m above the pipe and h_2 is unchanged.

It is the difference $\gamma_2 - \gamma_1$ that sets the sensitivity: a gage fluid close in specific weight to the flowing fluid gives a large, readable h_2 for a small pressure drop.

The inclined-tube manometer

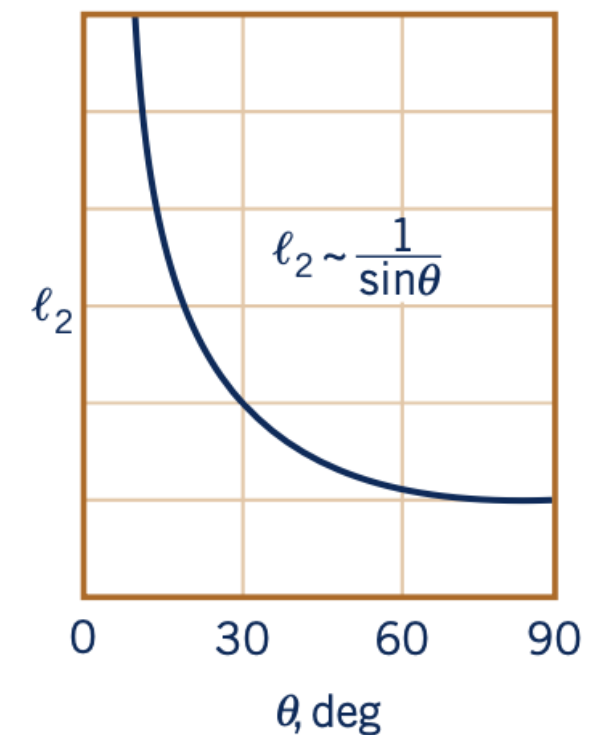
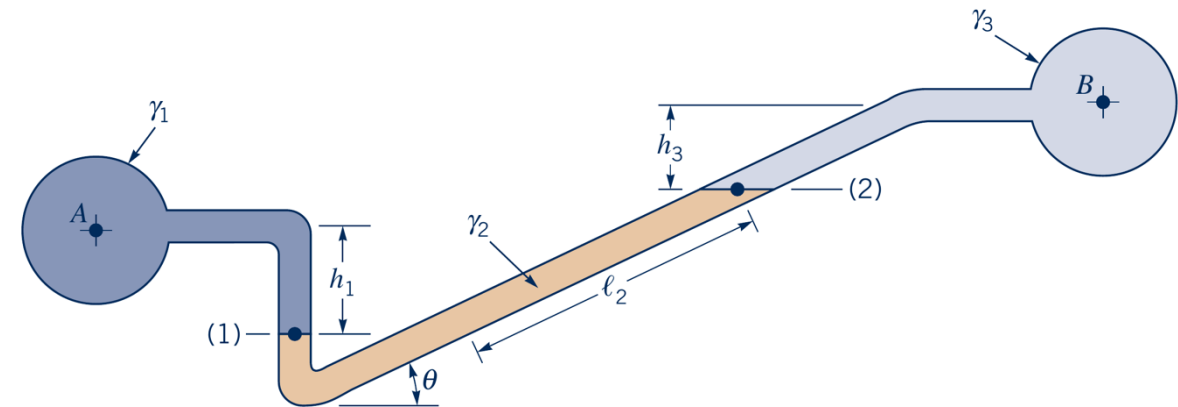
To read small pressure differences, one leg is inclined at an angle θ and the differential reading ℓ_2 is measured along the inclined tube. The vertical distance is $\ell_2 \sin \theta$:

$$p_A - p_B = \gamma_2 \ell_2 \sin \theta + \gamma_3 h_3 - \gamma_1 h_1$$

$$\frac{\ell_2}{h_2} = \frac{1}{\sin \theta}$$

If A and B contain a gas, the h_1 and h_3 terms drop out:

$$\ell_2 = \frac{p_A - p_B}{\gamma_2 \sin \theta}$$



The gain $1/\sin \theta$ diverges as $\theta \rightarrow 0$.

➡ **Inclined tubes measure small pressure differences accurately**

Practical notes on manometers

- **Capillarity is usually ignored:** in a simple U-tube with a meniscus in each leg the capillary effects cancel, provided the two tubes have the same diameter
- **Use wide bores**, about 13 mm or more, and the capillary rise becomes negligible anyway.
- **Water and mercury are the two common gage fluids:** both give a well-defined meniscus and both have well-known properties
- **The gage fluid must be immiscible** with every other fluid it touches
- **Temperature matters for accurate work**, because every specific weight in the equation depends on it

Manometers are accurate and need no calibration, but they are slow to read, unsuited to rapidly changing pressures, and impractical at high pressures.

Beyond manometers: the Bourdon gage

Many pressure measuring devices use the fact that **a force (pressure times area) causes an elastic object to deform.**



Bourdon gage.

The **Bourdon gage** is a very common type of pressure gage.

When the pressure within the hollow curved tube of the gage is increased, the tube tends to straighten.

Through a set of linkages, the resulting slight motion of the end of the tube is translated into the rotation of a dial that indicates the gage pressure.

Beyond manometers: the Bourdon gage

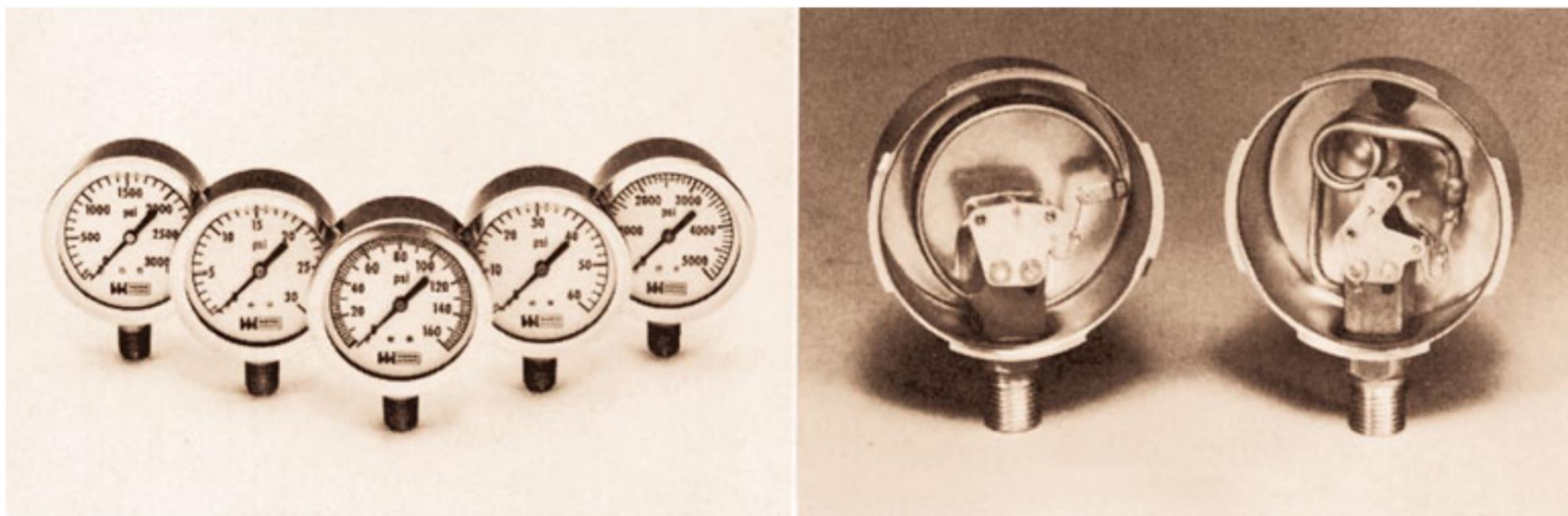
The deformation is small, but it drives a pointer on a dial.

The difference between the pressure inside the tube and the atmospheric pressure outside that moves it, so a **Bourdon gage** reads gage pressure, and reads zero at local atmospheric pressure.



Bourdon gage.

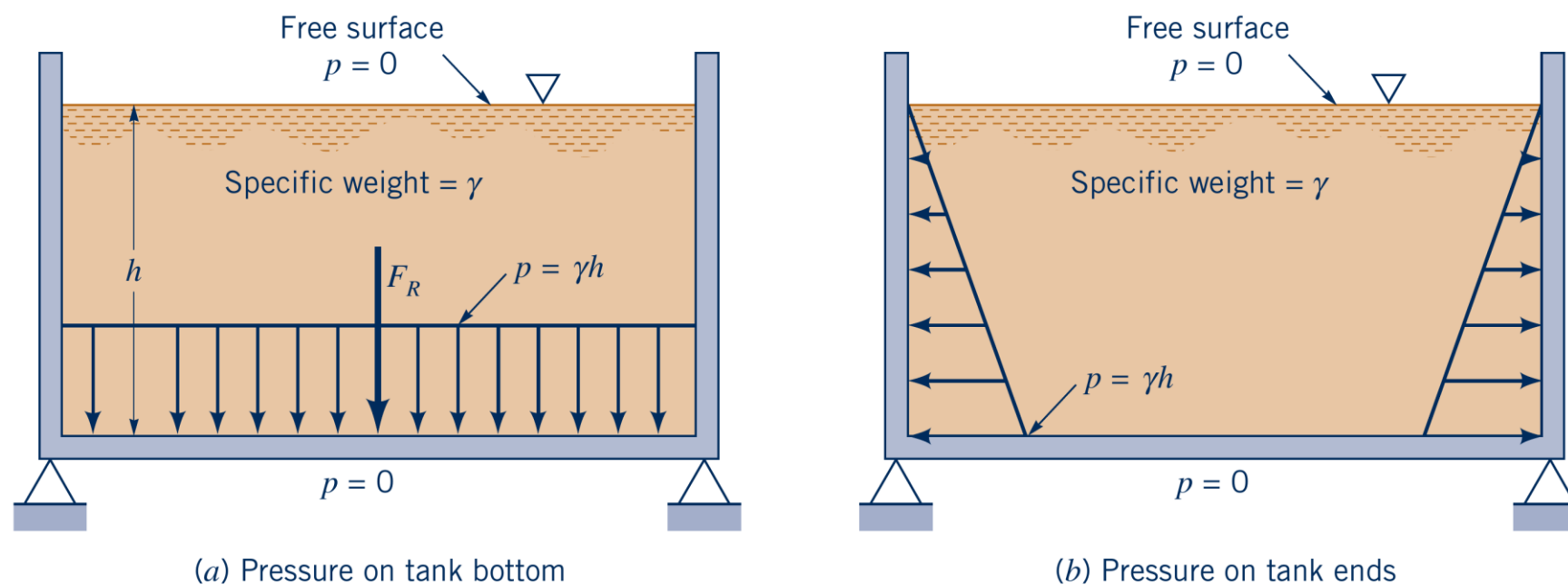
The **aneroid barometer** uses an evacuated element instead, and reads absolute pressure.



Liquid-filled Bourdon gages, and the tube inside them.

Hydrostatic force on a plane surface

When a surface is submerged, forces develop on it. Determining them is the design problem for storage tanks, ships, dams and hydraulic structures. In a fluid at rest the **force must be perpendicular to the surface**, since there are no shearing stresses, and the pressure varies linearly with depth.



■ **Figure 2.16** (a) Pressure distribution and resultant hydrostatic force on the bottom of an open tank. (b) Pressure distribution on the ends of an open tank.

On a horizontal surface, the tank bottom, the pressure is uniform and the resultant is simply p times the area, acting through the centroid:

$$\theta = 0 : \quad F_R = \gamma h A \quad \text{at the centroid}$$

On the ends the pressure is not uniform, and the resultant needs integration.

Atmospheric pressure acting on both sides cancels.

An inclined plane surface: the notation

Let the plane of the surface cut the free surface at O, at an angle θ .

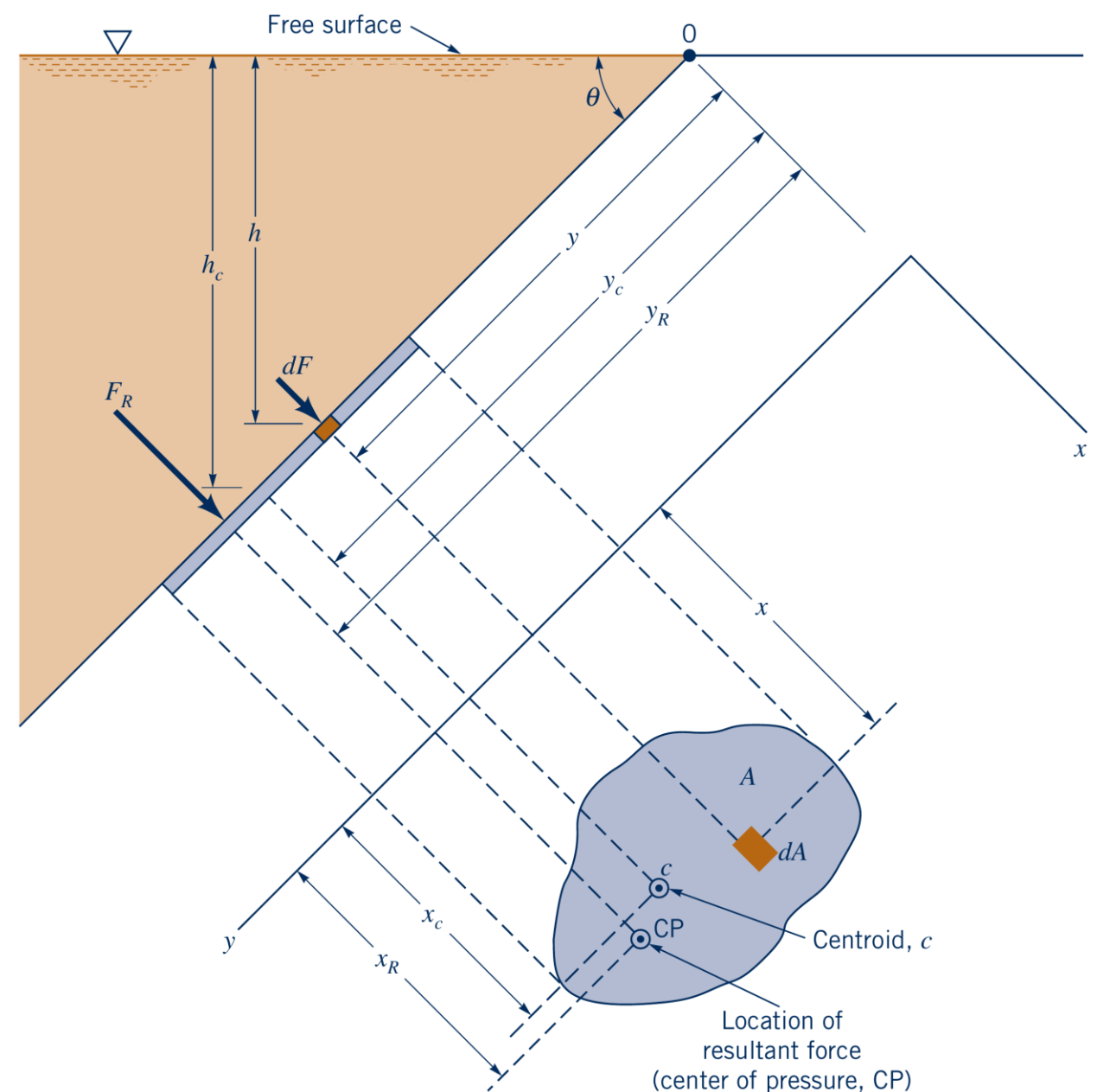
Take x–y in that plane with O as origin and the x axis along the free surface. The area may have any shape.

At depth h the force on the element dA is $dF = \gamma h dA$, perpendicular to the surface, and $h = y \sin \theta$. Summing over the area:

$$F_R = \int_A \gamma h dA = \int_A \gamma y \sin \theta dA$$

and with γ and θ constant:

$$F_R = \gamma \sin \theta \int_A y dA$$



■ **Figure 2.17** Notation for hydrostatic force on an inclined plane surface of arbitrary shape.

The magnitude of the resultant force

The integral is the **first moment of the area** about the x axis, so the centroid of the area is what appears:

$$\int_A y \, dA = y_c A$$

with that centroid measured from the x axis through O. Substituting,

$$F_R = \gamma A y_c \sin \theta$$

and since the depth of the centroid is that distance times $\sin \theta$,

$$F_R = \gamma h_c A$$

 **The pressure at the centroid, times the area**

The magnitude does not depend on θ : only on γ , on A , and on the depth of its centroid. **Since every dF is perpendicular to the surface, so is the resultant force.**

Where it acts: the centre of pressure

Intuition suggests the centroid, but that is wrong. Taking moments about the x axis:

$$F_R y_R = \int_A y dF = \gamma \sin \theta \int_A y^2 dA$$

so that, with the second moment of area about the x axis,

$$y_R = \frac{\int_A y^2 dA}{y_c A} = \frac{I_x}{y_c A}$$

$$y_R = \frac{I_{xc}}{y_c A} + y_c$$

and the **parallel-axis theorem**, about the centroidal axis:

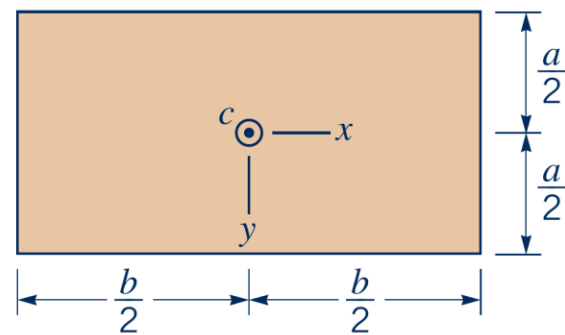
$$I_x = I_{xc} + A y_c^2$$

$$x_R = \frac{I_{xyc}}{y_c A} + x_c$$

The first term is positive, so **the centre of pressure lies below the centroid**, and moves towards it as the surface is sunk deeper.

If the area is symmetric about an axis parallel to y , the product of inertia vanishes and the resultant lies on the centroidal line.

Geometric properties of common shapes



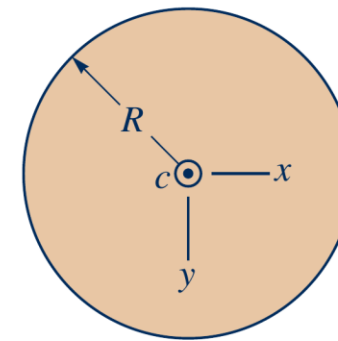
(a) Rectangle

$$A = ba$$

$$I_{xc} = \frac{1}{12} ba^3$$

$$I_{yc} = \frac{1}{12} ab^3$$

$$I_{xyc} = 0$$

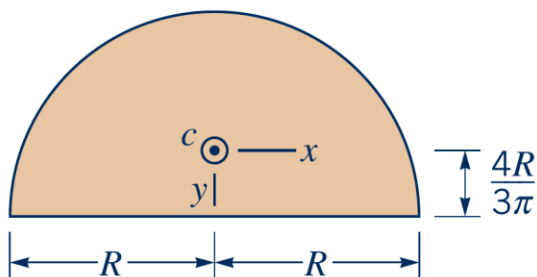


(b) Circle

$$A = \pi R^2$$

$$I_{xc} = I_{yc} = \frac{\pi R^4}{4}$$

$$I_{xyc} = 0$$



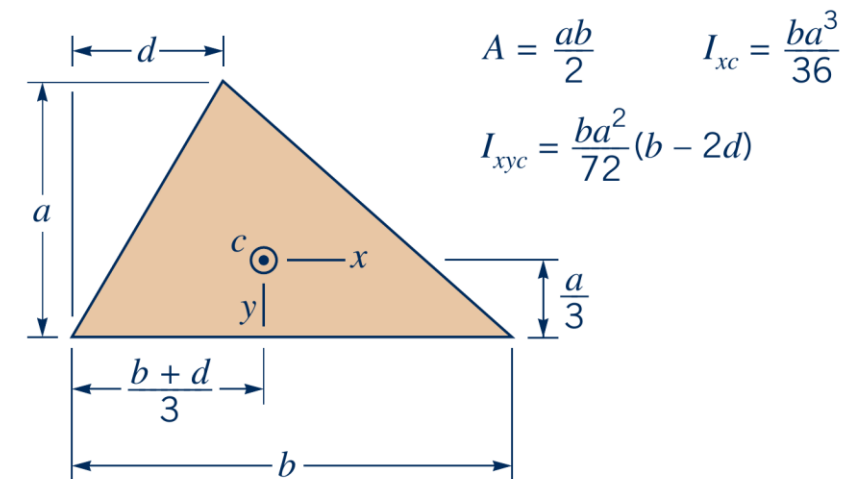
(c) Semicircle

$$A = \frac{\pi R^2}{2}$$

$$I_{xc} = 0.1098R^4$$

$$I_{yc} = 0.3927R^4$$

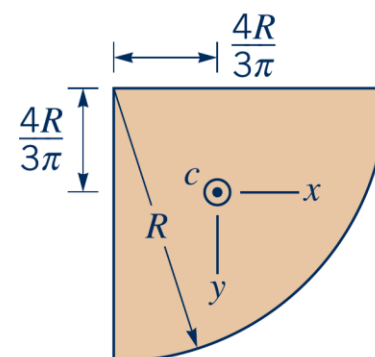
$$I_{xyc} = 0$$



(d) Triangle

$$A = \frac{ab}{2} \quad I_{xc} = \frac{ba^3}{36}$$

$$I_{xyc} = \frac{ba^2}{72}(b - 2d)$$



(e) Quarter circle

$$A = \frac{\pi R^2}{4}$$

$$I_{xc} = I_{yc} = 0.05488R^4$$

$$I_{xyc} = -0.01647R^4$$

■ **Figure 2.18** Geometric properties of some common shapes.

Centroid and centre of gravity

Two points appear in this lecture and they are easy to confuse:
one is pure **geometry**, the other depends on how the **weight is distributed**.

Centroid

$$y_c = \frac{\int y d\mathcal{V}}{\int d\mathcal{V}}$$

The first moment of volume over the volume. It depends on the shape of the body and on nothing else.

Centre of gravity

$$y_{CG} = \frac{\int y \gamma(y) d\mathcal{V}}{\int \gamma(y) d\mathcal{V}}$$

The first moment of weight over the weight, with $d\mathcal{W} = \gamma d\mathcal{V}$ for each element. It depends on how γ varies through the body.

If the specific weight is constant it comes out of both integrals and cancels:

$$y_{CG} = \frac{\int y \gamma d\mathcal{V}}{\int \gamma d\mathcal{V}} = \frac{\gamma \int y d\mathcal{V}}{\gamma \int d\mathcal{V}} = \frac{\int y d\mathcal{V}}{\int d\mathcal{V}} = y_c$$

For a body of uniform density the two coincide. In a stratified fluid the deeper elements weigh more, and they separate:

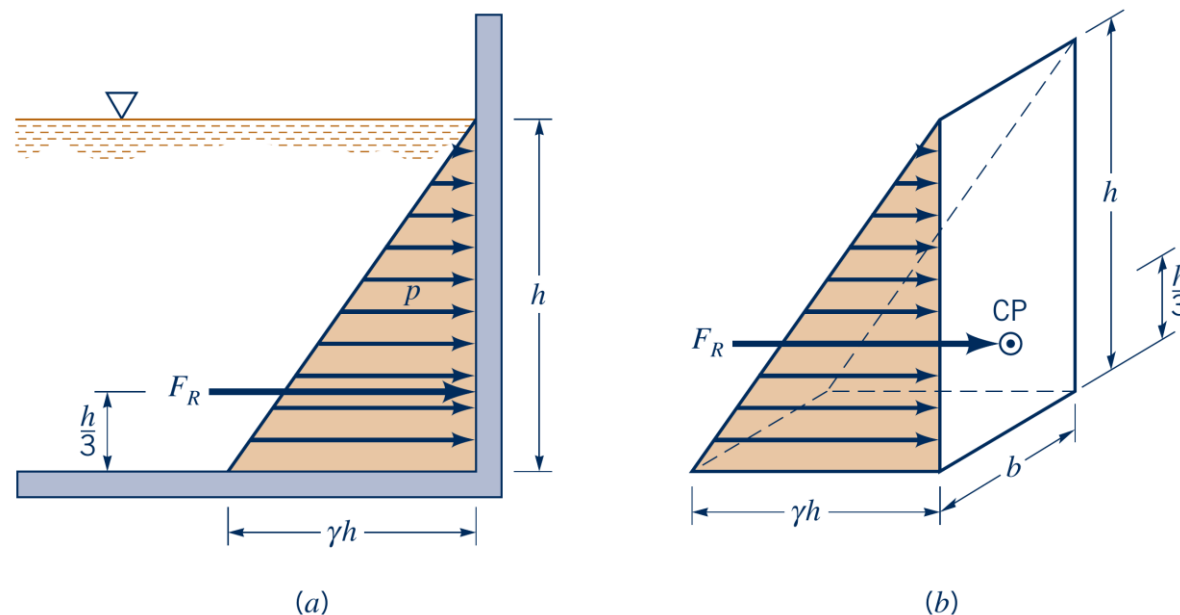
$$y_c = \frac{H}{2}, \quad y_{CG} > y_c$$

rectangle of height H , y downwards

The pressure prism

For a plane rectangular area there is a graphical reading of the same result. Along a vertical wall of constant width b the pressure runs from zero at the surface to γh at the bottom, so the average is at mid-depth:

$$F_R = p_{av} A = \gamma \left(\frac{h}{2} \right) A$$



■ **Figure 2.19** Pressure prism for vertical rectangular area.

In 3D the distribution is a solid whose base is the surface and whose height at each point is the pressure. Its volume is the force:

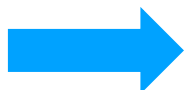
$$F_R = \text{volume} = \frac{1}{2}(\gamma h)(bh) = \gamma \left(\frac{h}{2} \right) A$$

The pressure prism

- **The magnitude of the resultant force** is the volume of the pressure prism
- **Its line of action** passes through the centroid of that volume
- **For the triangular prism** the centroid sits $h/3$ above the base, so the resultant acts one third of the height above the bottom of the wall

vertical rectangle $b \times h$: $F_R = \frac{1}{2}\gamma b h^2, \quad y_R = \frac{2}{3}h$

If the liquid surface is itself pressurised, the prism gains a rectangular block of height p_0 on top, and its volume is added.

 **Volume of the prism is the force, acting through its centroid**

Hoover Dam



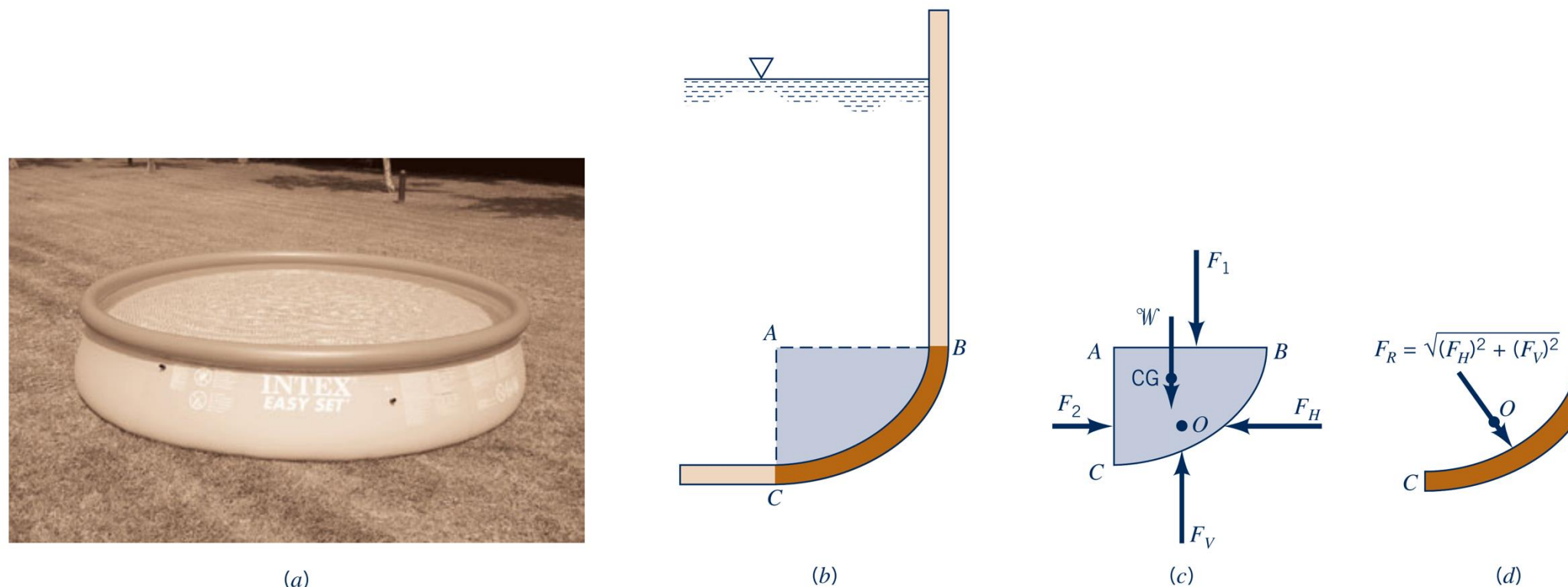
Video courtesy of U.S. Bureau of Reclamation

For an incompressible fluid at rest, the pressure increases linearly with the fluid depth. As a consequence, large forces can be developed on plane and curved surfaces in contact with the fluid.

Hoover dam, on the Colorado river, is the highest concrete arch-gravity type of dam in the US. The water behind Hoover dam is approximately 715 feet deep and at this depth the pressure is 310 psi. To withstand the large pressure forces on the face of the dam, its thickness varies from 45 feet at the top to 660 feet at the base.

Hydrostatic force on a curved surface

We have discussed plane surfaces only, but dams, pipes and tanks are curved. Integrating over a curved surface is tedious and gives no general formula. Instead, isolate the volume of fluid bounded by the surface and by its horizontal and vertical projections.



■ **Figure 2.23** Hydrostatic force on a curved surface. (Photograph courtesy of Intex Marketing, Ltd.)

F_1 and F_2 come from the plane-surface formulas; $\mathcal{W}$ is the weight of the enclosed fluid, acting through its centre of gravity. Equilibrium of the three-force system gives the horizontal and vertical components:

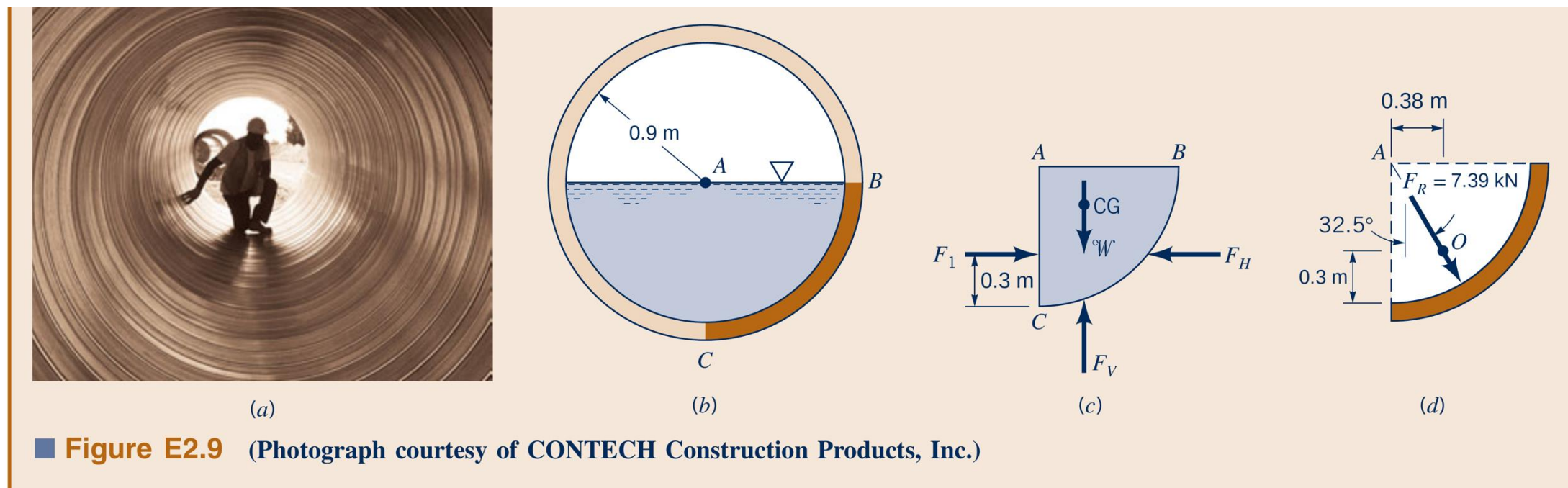
$$F_H = F_2, \quad F_V = F_1 + \mathcal{W}$$

and the resultant on the fluid is

$$F_R = \sqrt{(F_H)^2 + (F_V)^2}$$

Example — force on a curved conduit wall

A drainage conduit of diameter 1.8 m is half full of water at rest. Find the magnitude and line of action of the resultant force the water exerts on a 1 m length of the curved section BC.



$$F_1 = \gamma h_c A = (9.80)(0.45)(0.9 \times 1) = 3.97 \text{ kN}$$

$$W = \gamma V = (9.80) \left(\frac{\pi(0.9)^2}{4} \right) (1) = 6.23 \text{ kN}$$

The weight acts through the centroid of the quarter circle:

$$\bar{x}_{CG} = \frac{4R}{3\pi} = 0.382 \text{ m}$$

$$F_R = \sqrt{(3.97)^2 + (6.23)^2} = 7.39 \text{ kN} \quad \text{at } 32.5^\circ \text{ from the vertical}$$

The pop bottle



Video © B. R. Munson

The hydrostatic force distribution on a curved surface can be quite complex because the force acts normal to the surface, and the direction of the normal varies from point-to-point along the surface.

A pressure vessel must be designed to withstand the pressures generated.

If the weight of the pop is neglected, **the net force on the complex-shaped bottom of a pop bottle is equal to the pressure times the projected area, a circle the diameter equal to that of the bottle.**

Thus, when the pressure is 276 kPa, a force of approximately 2.6 kN is needed to fasten the bottom of a 109 mm diameter pop bottle to its cylindrical sides.

Lecture summary

The hydrostatic equations indicate that pressure depends on depth.

$$p = \gamma h$$

pressure-depth relation

$$p_A = \gamma_1 h_1$$

piezometer tube

$$p_A + \gamma_1 h_1 - \gamma_2 h_2 = 0$$

U-tube manometer

$$p_1 - p_2 = \gamma_2 \ell_2 \sin \theta$$

inclined tube

$$F_R = \gamma h_c A$$

force on a plane surface

$$y_R = \frac{I_{xc}}{y_c A} + y_c$$

centre of pressure

$$F_R = \frac{1}{2} \gamma h^2 b$$

the pressure prism

$$F_R = \sqrt{F_H^2 + F_V^2}$$

curved surfaces